

Project Report

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Real-World Application of Time Value of Money (TVM) in EV Car Financing

1. Introduction

1.1 Project Objective

The primary objective of this project is to analyze the real-world application of Time Value of Money (TVM) in Electric Vehicle (EV) financing. While TVM is usually taught theoretically, this study focuses on how interest rates, loan tenure, and repayment structure affect the actual cost of borrowing in EV purchases.

1.2 Importance of TVM in Personal Finance

TVM states that money today is worth more than the same amount in the future due to its earning capacity. In car financing, TVM enables consumers to understand:

- The difference between principal amount and total repayment.
- How compounding affects monthly installments (EMIs).
- How foreclosure charges and prepayment decisions impact overall cost.

2. Data Collection (Evidence)

To maintain authenticity, our group visited Oberai Motors (Authorized Tata Motors Dealer) located on Saharanpur Road, Dehradun. We collected official price quotations for two EVs from different segments.

2.1 Vehicle Details

Parameter	Vehicle 1 (Entry EV)	Vehicle 2 (Premium EV)
Model	Tata Tiago EV (XT 19.2 kWh) Tata Harrier EV (Adventure 65)	
Quotation Date	05 Sept 2025	28 Jun 2025
Ex-Showroom Price	₹ 8,99,000	₹ 21,49,000
Insurance	₹ 45,574	₹ 1,24,057
Registration	₹ 6,000	₹ 8,000
Other Charges	₹ 2,862 (Add-ons)	₹ 22,190 (TCS/Fastag)
Total On-Road Price	₹ 9,53,436	₹ 23,03,247

(Scanned copies of the quotations)

**Oberai Motors**
P.O. Majra Saharanpur Road Dehradun-248001
Tel: 2645011 to 2645013 Fax: 0135- 2645019

For Sale Comfort: 908411113,
For Finance Comfort: 9639016031,

DATE : 05 SEPT 2025

		TIAGO EV									
Variant		ESP	Insurance	Registration	TCS	PANIC BUTTON	On Road Price	ADD ON	EW (125000 OR UP TO 1 YEAR ADD.)	TMGA ESSENTIAL PACK	On Road Price
TIAGO EV	Tiago EV XE 19.2 KWH (3.3KW AC)	799000	45574	6000		0	850574	6398	17999	12000	886971
	Tiago EV XT 19.2 KWH (3.3KW AC)	899000	48436	6000		0	953436	6398	17999	12000	989833
	Tiago EV LR XT 24 KWH (3.3KW AC)	1014000	51727	6000	10140	0	1081867	6398	17999	12000	1118264
	Tiago EV LR XZ+ TECH LUX 24 KWH (3.3KW AC)	1114000	54589	6000	11140	0	1185729	6398	17999	12000	1222126

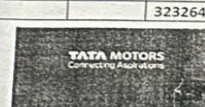
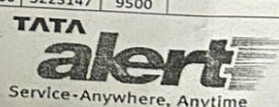
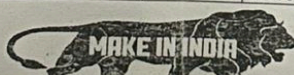


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DATE : 28 JUN 2025

		NEW HARRIER EV											
Variant		ESP	Insurance	Registration	TCS	Fast ag	On Road Price	Insurance Add ON	AMC (30000 OR UP TO 2 YEAR)	EW 2 YEAR unlimited Km	TMGA Essential Pack	On Road Price	
NEW HARRIER EV	HARRIER EV ADVENTURE 65	2,149,000	124057	8000	21490	700	2303247	9500				231274	
	HARRIER EV ADVENTURE 65 ACFC	2,198,000	126140	8000	21980	700	2354820	9500				236437	
	HARRIER EV ADVENTURE S 65	2,199,000	126182	8000	21990	700	2355872	9500				236537	
	HARRIER EV ADVENTURE S 65 ACFC	2,248,000	128266	8000	22480	700	2407446	9500				241694	
	HARRIER EV FEARLESS+ 65	2,399,000	132686	8000	23990	700	2564376	9500				257387	
	HARRIER EV FEARLESS+ 65 ACFC	2,448,000	136789	8000	24480	700	2617969	9500				262746	
	HARRIER EV FEARLESS+ 75	2,499,000	138939	8000	24990	700	2671629	9500				268117	
	HARRIER EV FEARLESS+ 75 ACFC	2,548,000	141020	8000	25480	700	2723200	9500				273270	
	HARRIER EV EMPOWERED 75	2,749,000	149568	8000	27490	700	2934758	9500				294425	
	HARRIER EV EMPOWERED 75 ACFC	2,798,000	151652	8000	27980	700	2986332	9500				299583	
	HARRIER EV EMPOWERED ST 75	2,824,000	152758	8000	28240	700	3013698	9500				302319	
	HARRIER EV EMPOWERED ST 75 ACFC	2,873,000	154842	8000	28730	700	3065272	9500				307477	
	HARRIER EV EMPOWERED AWD 75	2,899,000	155948	8000	28990	700	3092638	9500				310213	
	HARRIER EV EMPOWERED AWD75ACFC	2,948,000	158029	8000	29480	700	3144209	9500				315370	
	HARRIER EV EMPOWERED AWD ST 75	2,974,000	159133	8000	29740	700	3171573	9500				318107	
	HARRIER EV EMPOWERED AWDST75FC	3,023,000	161217	8000	30230	700	3223147	9500				323264	



3. Loan Structure & Assumptions

Based on the dealership's financed department, we assumed:

- Down Payment: 20% of On-Road Price
- Loan Amount: 80% of On-Road Price
- Interest Rate (r): 9.25% p.a. (reducing balance)
- Loan Tenure (n): 5 years (60 months)
- Processing Fees: Waived

3.1 Principal Loan Amount Calculation

A. Tata Tiago EV

$$\text{Principal} = 9,53,436 - (20\% \times 9,53,436) = 7,62,749$$

B. Tata Harrier EV

$$\text{Principal} = 23,03,247 - (20\% \times 23,03,247) = 18,42,598$$

4. EMI Calculation (TVM Application)

EMI is calculated using the formula derived from the PV of an annuity:

$$E = P \times r \times \frac{(1+r)^n}{(1+r)^n - 1}$$

Where:

- E = EMI
- P = Principal
- r = Monthly Interest Rate
- n = Number of months

Input Variables

$$r = \frac{9.25}{12 \times 100} = 0.0077083$$
$$n = 60$$

4.2 EMI for Tata Tiago EV (P = 7,62,749)

$$E = 7,62,749 \times 0.0077083 \times \frac{1.585}{0.585}$$

EMI = ₹ 15,925

4.3 EMI for Tata Harrier EV (P = 18,42,598)

EMI = ₹ 38,469

5. Foreclosure & Repayment Analysis

Scenario:

The Tiago EV loan is foreclosed after 24 months (2 years).

5.1 Outstanding Principal at Month 24

$$B = E \times \frac{1 - (1 + r)^{-(n-k)}}{r}$$
$$B = 15,925 \times \frac{1 - (1.007708)^{-36}}{0.007708}$$

Outstanding Balance = ₹ 4,96,412

5.2 Foreclosure Charges

$$\text{Penalty} = 4\% + \text{GST (18\%)} = 4.72\%$$

$$\text{Penalty} = 4,96,412 \times 4.72\% = ₹23,430$$

5.3 Financial Decision Matrix

Option A: Foreclose Now Option B: Continue EMIs

Pay Outstanding: ₹ 4,96,412 Pay 36 EMIs: ₹ 5,73,300

Pay Penalty: ₹ 23,430 —

Total: ₹ 5,19,842 Total: ₹ 5,73,300

Conclusion:

The customer saves ₹ 53,458 by foreclosing early.

6. TVM Interpretation & Insights

6.1 Total Cost of Borrowing (Harrier EV)

$$38,469 \times 60 = 23,08,140$$

$$\text{Interest Paid} = 23,08,140 - 18,42,598 = ₹4,65,542$$

The customer pays ₹ 4.65 lakh extra due to the time value of money.

6.2 Flat Rate vs. Reducing Rate Misconception

- 6% Flat Rate EMI $\approx$ ₹ 39,922
- 9.25% Reducing Rate EMI $\approx$ ₹ 38,469

Even though 6% appears lower, the flat rate method charges interest on the full principal for all 5 years → making it costlier.

7. Group Learning & Conclusion

7.1 Key Learnings

- On-road prices include many hidden components (Insurance, TCS, accessories).
- EMI reduces when tenure increases, but total interest increases sharply.
- TVM formulas match real bank amortization tables.
- Foreclosure is financially beneficial if done after sufficient interest has already been paid.

7.2 Mapping with Course Outcomes

Course Outcome	Achievement
CO1 – Data Collection	Achieved via dealership visit
CO2 – EMI Application	Correct use of annuity formula
CO3 – Foreclosure Analysis	Calculated outstanding balance & penalty
CO4 – Decision Making	Compared rates & evaluated financial impact

Appendix: Evidence of Authenticity

Visiting card



Handwritten rough notes taken at outlet

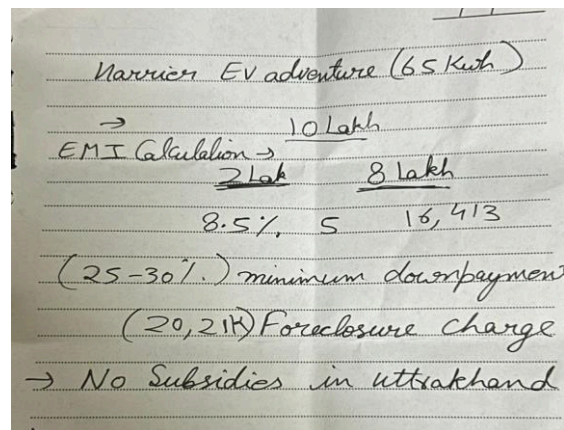
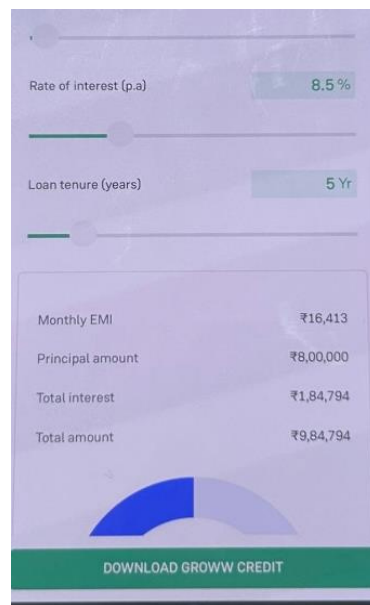


Photo of EMI calculation

Handwritten rough notes taken at outlet



Other Proofs

